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Milad Akrami, Ehsan Jamshidpour, Babak Nahid-Mobarakeh, Serge Pierfederici, Vincent Frick. Sensorless Control Methods for BLDC Motor Drives: A Review. IEEE Transactions on Transportation Electrification, 2024, 11 (1), pp.135-152. <10.1109/TTE.2024.3387371>. <hal-04559983>

HAL Id: hal-04559983

<https://hal.univ-lorraine.fr/hal-04559983v1>

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Sensorless Control Methods for BLDC Motor Drives: A Review

Milad Akrami, Ehsan Jamshidpour, *Senior Member, IEEE*, Babak Nahid-Mobarakeh, *Fellow, IEEE*, Serge Pierfederici, and Vincent Frick

Abstract—Permanent magnet motors, such as Brushless DC (BLDC) motors, are widely adopted in today's industry for their high power density, torque-to-weight ratio, and uncomplicated design. Furthermore, electrical motor drive systems are increasingly adopting (position) sensorless control methods. Sensorless techniques have several advantages over sensor-based methods, including enhanced reliability, prolonged motor lifespans, simplified inverter-motor connections, and lower costs. Several sensorless techniques have been proposed over the past few decades, such as methods based on terminal voltage measurement, third harmonic back electromotive force (back-EMF) signals, and estimation methods. Research in recent years has focused on improving conventional sensorless methods and developing new ones, including those based on flux linkage functions and neural networks. These advancements address challenges associated with high-speed BLDC drives where the phase lag of the LPF is significant, tackle issues in very low-speed applications where back-EMF signal amplitudes are low, improve position estimation speed to enable one-cycle estimation, enhance control system accuracy for BLDC motors with nonideal back-EMFs, and consider asymmetric back-EMF signals. This paper examines the benefits, limitations, and challenges of sensorless drives for BLDC motors, alongside innovative solutions for nonideal and asymmetric back-EMF issues, with an eye toward future developments.

Index Terms—Brushless DC motor, Sensorless Control, Non-ideal back-EMF, Asymmetric back-EMF.

I. INTRODUCTION

Variable-speed drives have become popular in modern industries. As a result of this growth, PM motor drives with variable speeds have also become more popular in various applications [1]–[4]. Thanks to their several features such as high power density, high torque-per-weight ratio, and simple structure, BLDC motors have been widely used in the aerospace, industrial controls, and EV industries [5]–[11].

Information of the rotor position and speed is necessary for effective drive of the BLDC motor. Various mechanical position sensor schemes, such as hall sensors, photoelectric encoders, and rotary transformers are commonly used for this purpose [12]–[14]. However, traditional position sensors have certain drawbacks. Using position sensors not only increases the size and inertia of the motor but also reduces its power density and increases the complexity and cost of the drive system [15]. In addition, these sensors are vulnerable to temperature variations and interference with electromagnetic noise [15]. Sensor mounting accuracy affects motor performance, particularly for motors with a higher number of poles [16]. Therefore, sensorless control methods have gained significant attention in the current literature as effective solutions to the

forementioned problems [15], [17]. Sensorless control methods, which use voltage and current measurements to determine real-time position of the rotor, are being utilized more and more in the electrical motor drive systems. There are several advantages to sensorless control methods over sensor-based ones, including improved reliability, longer motor service life, simpler inverter-motor connections, and lower system costs [18], [19].

In this paper, a comprehensive review on different sensorless control methods for BLDC motors is conducted. The methods are classified based on the way they work. Their advantages and issues will be discussed and compared. The paper is organized as follows: in section II, the working principle of BLDC motor and its model will be discussed. In Section III, the general BLDC motor drive system will be presented. Section IV discusses different sensorless control methods for BLDC motors. Recent studies regarding the dealing with the issues of sensorless control of BLDC motors with nonideal and asymmetric back-EMF is discussed in Section V and VI, respectively. Conclusion is given in Section VII.

II. BLDC MOTOR

A three-phase PM motor is similar to an induction motor where a three-phase stator winding is designed to create an air-gap flux distribution, either trapezoidal or sinusoidal, depending on whether it's a BLDC motor or a sinusoidal wave-wound synchronous motor. Fig. 1 shows a simple two-pole motor with a 180-degree magnetic pole arc and a three-phase stator winding featuring two slots for each phase (the mechanical phase angle between two coils, δ , is equal to 30° in this case).

As long as there is no fringing, the magnetic flux in an air gap behaves like a square wave in a BLDC motor, as depicted in Fig. 2. It should be noticed that the actual phase back-EMF is trapezoidal due to magnetic fringing and motor construction. The rotation of the rotor generates a voltage in the stator windings and the position of the rotor changes the strength of the magnetic field linearly. The highest positive and negative magnetic field strengths occur at 0 degrees and 180 degrees, respectively. In addition, during each switching period of a BLDC motor, only two phases of a star-connected motor conduct electricity and the other phase remains unexcited [20].

A. BLDC motor model

The basic torque and voltage equations of BLDC motors are similar to those of DC brushed motors [21]. Fig. 3 illustrates

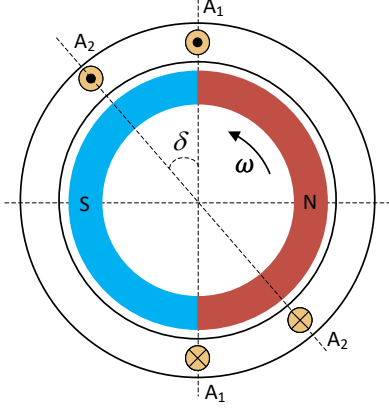


Fig. 1. Cross-section of an ideal BLDC motor with two coils in one phase (A_1 and A_2) [20].

a block diagram of a typical drive system used for a BLDC motor. The BLDC motor model can be represented as:

$$V = Ri + \frac{d}{dt}[L(\theta)i + \lambda_M] = Ri + L(\theta)\frac{di}{dt} + i\omega\frac{d}{d\theta}L(\theta) + e \quad (1)$$

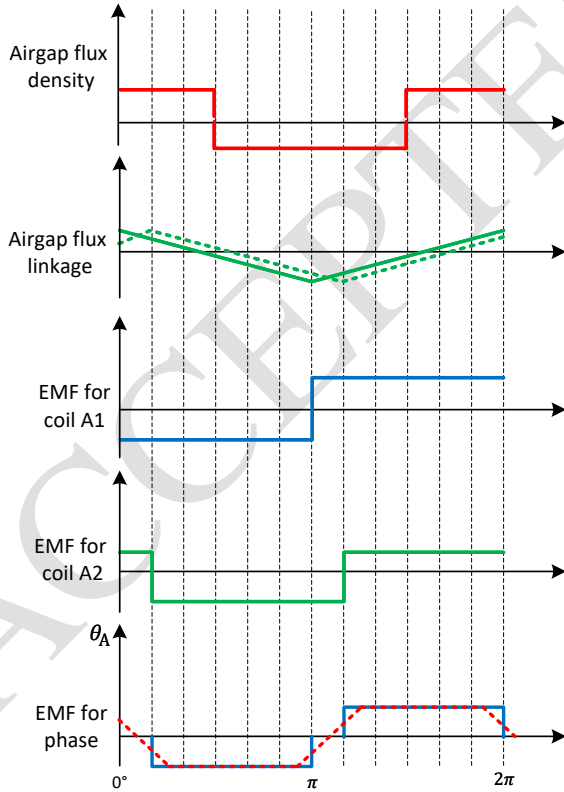


Fig. 2. Waveforms for an ideal BLDC motor including airgap flux density, airgap flux linkage, and EMF for each coil and one phase [20].

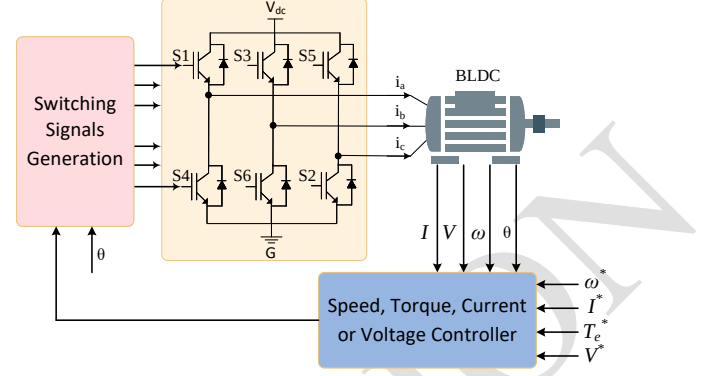


Fig. 3. Structure of the general BLDC motor drive featuring a three-phase inverter [17].

where R and L are stator impedance and inductance. Also, V , i , and e are motor phase terminal voltage, current, and Back-EMF, respectively. Flux caused by permanent magnets is represented by λ_M . Parameters θ and ω are rotor phase angle and angular velocity.

It should be noted that $i\omega\frac{d}{d\theta}L(\theta)$ is zero as there is no change in inductance according to the position. Therefore, the model is simplified as:

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \quad (2)$$

where i_a, i_b , and i_c are phase currents, V_a, V_b , and V_c are phase voltages, and e_a, e_b , and e_c are phase back-EMF voltages. It should be mentioned that $L = L_s - M$, that L_s is the stator self-inductance and M is the mutual inductance between two phases. As $i_a + i_b + i_c = 0$, the inductance matrix is simplified as described in (3).

$$\begin{bmatrix} L_s & M & M \\ M & L_s & M \\ M & M & L_s \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (3)$$

The electromagnetic torque of the BLDC motor is described as:

$$T_e = \frac{1}{\omega}(e_a i_a + e_b i_b + e_c i_c) \quad (4)$$

Assuming that there is no phase difference between the current and back-EMF, the electromagnetic torque is equal to

$$T_e = \frac{2EI}{\omega} \quad (5)$$

where E and I are back-EMF voltage peak amplitude and DC-link current amplitude.

In addition electromagnetic torque can be presented as:

$$T_e = T_L + J\frac{d\omega}{dt} + B\omega \quad (6)$$

where, T_L is load torque, J is inertia, and B is the viscous damping.

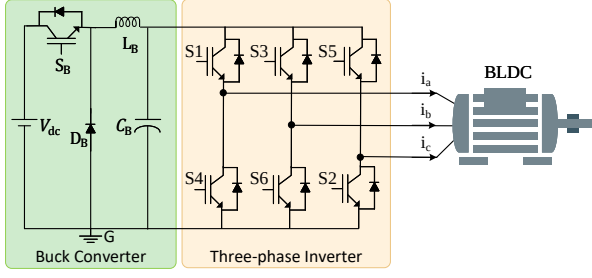


Fig. 4. Schematic of the PAM inverter topology.

III. BLDC MOTOR DRIVE

Specific control strategies are employed by the drive system in order to control various motor parameters such as speed, torque, or current [17]. Based on the rotor's position within every 60 electrical degrees, these strategies generate switching signals that determine the selection of a pair of switches for the three-phase inverter.

Pulse Amplitude Modulation (PAM) control has been introduced to improve the drive of BLDC motor for wide-speed range and particularly low-speed applications. Fig. 4 illustrates the PAM drive topology for BLDC motors. It consists of a buck converter and a three-phase inverter. The buck converter regulates the DC-link voltage of the three-phase inverter [22]–[25]. Compared to conventional BLDC motor drives based on pulse width modulation (PWM) drive technique, PAM drive method of BLDC motors offer more efficiency and lower current total harmonic distortion (THD) [26], [27]. Utilizing the PAM technique, the inverter operates at the motor's fundamental frequency, which effectively decreases switching losses and removes the demand for high-bandwidth current controller [27]. However, two switches and one inductor should be added, which require additional hardware, resulting in lower reliability and higher costs. An inverter control circuit with a buck converter in front of the three-phase inverter was proposed to reduce those losses, particularly for small inductance motors [28]. With the buck converter adjusting the DC-link voltage, the three-phase bridge performs only commutation without modulation.

A comparison of PWM and PAM drive method of BLDC motors have been conducted in [29]. When the PAM technique is employed, there is no PWM applied to phase voltages, which eliminates the need for low-pass filters (LPFs) to mitigate PWM harmonics. Consequently, voltage delay can be reduced, which is particularly crucial at high speeds. Therefore, the back-EMF estimation method is enhanced within high-speed ranges [30]. Another PAM sensorless control method for BLDC motors is presented in [30], as depicted in Fig. 5. To improve the performance of the control system, a method for compensating the voltage spike in low-speed range and a method for compensating the phase delay in high-speed range is proposed in [30]. Also, a combination of PWM and PAM control method has been proposed in [31] to reduce the DC-

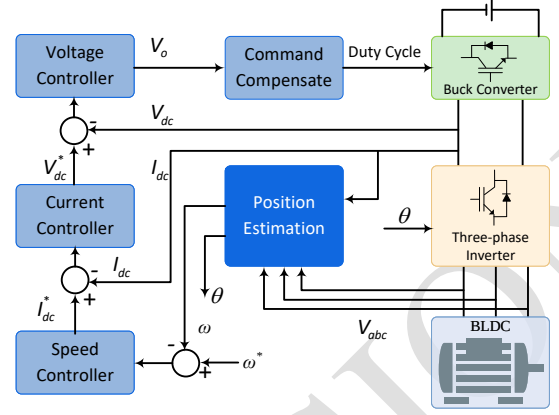


Fig. 5. Sensorless BLDC drive with DC-link voltage regulated PAM inverter [30].

link current ripples comparing to the conventional PWM drive method.

IV. SENSORLESS CONTROL METHODS FOR BLDC MOTORS

Sensorless control is a method for controlling BLDC motors without using external position sensors [32]. As an alternative to traditional position and speed sensors, sensorless control methods use a variety of techniques in order to estimate the rotor's position and commutation timing [33]. In order to determine the rotor's position, these techniques often monitor motor current, voltage, or back-EMF signals [34], [35]. A sensorless control system reduces complexity, improves reliability, and reduces costs by eliminating external position sensors. Automotive, robotics, and home appliances are just some of the industries where PM motor control using sensorless techniques can be found [36], [37].

A review of the structure, design, modeling and drive of BLDC motors has been conducted in [38]. The paper does not, however, provide a comprehensive overview of the control methods for BLDC motors and in particular the sensorless methods. Another overview of sensorless control for BLDC motors is presented in [39]. Nevertheless, it does not address the latest developments in this domain, particularly issues related to BLDC motors with nonideal and asymmetrical back-EMFs.

The sensorless control methods for BLDC motors can be classified into different types based on their operational methods, as discussed in [39]:

- **Back-EMF detection-based methods:** This category can be further divided into direct back-EMF detection and indirect back-EMF detection. Direct back-EMF detection methods typically rely on terminal voltage sensing, while indirect methods include those based on third harmonic back-EMF, back-EMF integration, and approaches using freewheeling diode current measurement.
- **Estimation and model-based methods:** These methods utilize the mathematical model of the BLDC motor

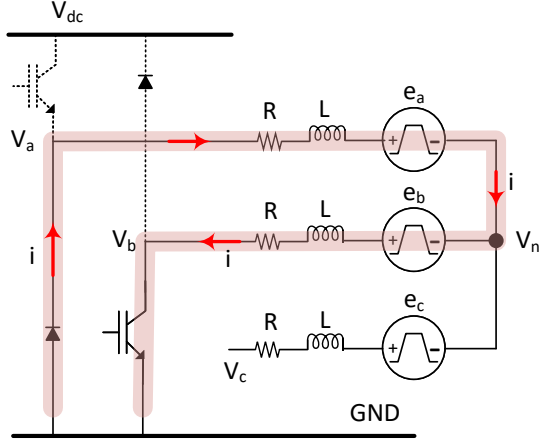


Fig. 6. Back-EMF detection in the PWM off-time moment [44].

drive system. They can be categorized into observer-based methods, flux linkage function-based methods, and stochastic filters, such as the Kalman filter (KF).

- Fuzzy control and artificial intelligence (AI)-based methods: Fuzzy control methods involve using linguistic variables to represent and control system behavior, while AI-based methods may include neural networks, machine learning algorithms, and other artificial intelligence approaches.

These methods will be further described and investigated in the following subsections.

A. Sensorless control methods based on zero-crossing point (ZCP) approach (Back-EMF based)

It is common to use model-based techniques for medium- and high-speed applications. Conventional methods, such as back-EMF and terminal voltage measuring, are among the most commonly used [40]–[42]. An automotive fuel pump BLDC motor drive based on microcontrollers is presented in [43]–[45]. The back-EMF signal is obtained by measuring the voltage of the unexcited phase of the motor, as shown in Fig. 6.

Voltage expressions when phase C is unexcited and current flows through phases A and B in the converter can be determined by ignoring the forward voltage of switches and diodes:

$$\begin{cases} V_n = -Ri - L\frac{di}{dt} - e_a \\ V_n = +Ri + L\frac{di}{dt} - e_b \end{cases} \quad (7)$$

which parameter V_n is the neutral point voltage. Equation (7) leads to:

$$V_n = -\frac{e_a + e_b}{2} \quad (8)$$

Assuming the motor phases to be balanced ($e_a + e_b + e_c = 0$), the terminal voltage V_c is equal to:

$$V_c = e_c + V_n = \frac{3}{2}e_c \quad (9)$$

Accordingly, the measured terminal voltage for the unexcited phase is proportional to the back-EMF voltage of the corresponding phase. It is possible to generate commutation signals from ZCPs of the back-EMF voltage, which are identical to those of the terminal voltage in the unexcited phase [44], [46]–[49]. As explained in [50], the previous method required a minimum interval of PWM off time. In this case, the switches that are used as motor drives must be operated at a duty cycle below 100%. Previous method cannot be used in automotive applications, as they require full-duty cycle operation. There is an improved method described in [50] for detecting back-EMF ZCPs during PWM on time. Detecting back-EMF in low-speed applications employs an off-time method while detecting it in high-speed applications employs an on-time method.

A method to detect the commutation points of a BLDC motor was described in [51] using symmetric terminal voltages of the unexcited phase. In [52], a four-switch position sensorless method is described for BLDC motor drives. This paper presents a sensorless control scheme for three-phase BLDC motors using field-programmable gate array (FPGA) technology. The authors in [53] have investigated BLDC motor safety operation using line-to-line voltages sensorless control techniques for ZCP detection. The paper shows that the PWM pattern affects sensorless control performance. Based on the parameters provided, it is shown whether the motor can operate at predetermined torques and speeds theoretically.

Sensorless methods based on terminal voltage measurements fail to provide reliable results when motor speeds are near zero, as the back-EMF is very small at low speed range [54], [55]. In [56], a LPF with variable cutoff frequency was developed to reduce the disturbance of the line-to-line voltages and to improve the drive performance of the BLDC motor. This method deals with the problem of severe phase delay due to low-speed range and signal distortion at higher speeds. Therefore, the cutoff frequency of LPF increases when the speed increases and vice versa. The transfer function of LPF with variable cutoff frequency can be described as [56]:

$$H_{LPF} = \frac{2\pi f_c}{s + 2\pi f_c} \quad (10)$$

where f_c is the cutoff frequency of the filter, which is expressed as $f_c = k_c f$. Parameter k_c is a proportional coefficient between 0 and 1 and f is the frequency of input signal of the LPF.

Fig. 7 summarizes the compensation approach, which includes an open-loop approach to compensate for the LPF phase delay and a closed-loop approach to compensate for uncertainty commutation errors. When this LPF is employed, a phase lag is created that must be compensated for, and in high-speed applications, this lag can be significant [57], [58].

ZCPs can be detected in BLDC motors using virtual line voltage as a sensorless method [59]. In [60], a hybrid sensorless driving method using virtual neutral voltage is proposed for high-speed BLDC motors. There is no discussion of the system's performance at high speeds, however. In [61], the virtual neutral voltage is proposed as another sensorless

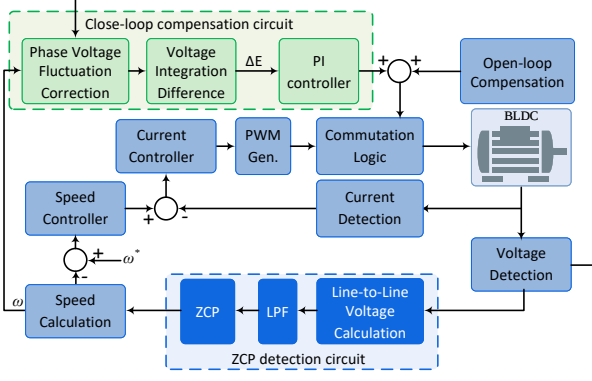


Fig. 7. Sensorless control method proposing open-loop and close-loop compensation blocks [56].

hybrid control method for BLDC motors. It is difficult to produce commutation signals from voltage signals as a result of commutation notches and electromagnetic interference. As a result, a low-pass filter is required for this method.

To overcome the phase delay caused by LPFs, Fig. 8 illustrates a sensorless control method for PMSMs that relies on transitions between " $90 - \alpha$ " and " $150 - \alpha$ " commutation mode, where α is the commutation retarding.

If α is below 90 electrical degrees, Z_1 will be shifted to Q_1 before M_1 . To ensure precise commutation signaling, Q_1 should be advanced by $90 - \alpha$ (where $90 - \alpha > 0$ must be met) electrical degrees to reach the exact commutation point M_1 . In medium- and high-speed range applications, the LPF results in phase lag bigger than 90 electrical degrees. Therefore, using the " $90 - \alpha$ " commutation mode will make the system unstable. To achieve the " $150 - \alpha$ " mode, Q_1 must be shifted by $150 - \alpha$ (where $150 - \alpha > 0$) electrical degrees to reach the specific commutation point N_1 . However, this method suffers from a conductivity error [57].

In [62], an approach is presented for reducing the commutation torque ripple of BLDC motor drives in air conditioners. However, the proposed method does not control the current in the stator. Fig. 9 illustrates the position sensorless control scheme proposed in [63], [64] for BLDC motors used in Photovoltaics (PV)-based water pump applications. An LPF is used to detect back-EMF signals with this method, which is based on line voltage measurements.

A sensorless control system for BLDC motors is developed for an automobile fuel pump in [65]. They propose a sensorless approach based on a hysteresis comparator and a high-torque starting mechanism. As shown in Fig. 10, the hysteresis comparator compensates for phase delays caused by LPF and prevents ripples in voltage caused by noise and transient output transitions. A specially designed inverter circuit is used in [66] to demonstrate a Fourier-series method for compensating phase delays of ZCPs detection.

A method of compensating the commutation shift of sensorless BLDC motors based on virtual Hall signals is presented in [67], [68]. As part of this technique, the commutation error is determined by integrating both the line-voltage difference

integral across a 60-degree electrical span as well as phase current just before commutation. An alternative sensorless motor control method for compensating for commutation errors is shown in Fig. 11. Zero-crossing filtered signals must be delayed by $90 - \alpha$ electrical degrees for them to be commutated without compensation; where α indicates the phase delay caused by the LPF, hardware, and software. By utilizing commutation freewheeling signals and ZCP signals, a real-time compensation time is calculated [69].

There are several limitations to methods measuring back-EMF and terminal voltages: they do not perform well at low speeds because of the small magnitude of the back-EMF, they do not perform well during speed variations caused by the LPF, and they have discrete speed estimation, resulting in difficulties in closed-loop speed control with significant steady-state speed variations within the low to medium-speed range.

B. Sensorless control methods based on third harmonic approach (Back-EMF based)

The third harmonic of the back-EMF voltage is utilized for sensorless control of BLDC motors. Compared to the previous technique, this method addresses the problems associated with the ZCP detection approach [70]–[72]. The third harmonic does not suffer from PWM noise, unlike the back-EMF, which is affected by it. Therefore, a higher low-pass filter cutoff frequency can be achieved. The third harmonic of the back-EMF is used in the sensorless commutation control system to maintain consistent phase relationships regardless of motor speed [70]. However, the application of this method at low speeds is limited by significant noise interference and integral errors.

A third harmonic back-EMF combined with phase-locked loops (PLLs) is another approach proposed in [72] for sensorless control of PM brushless motors. Additionally, [73] presents a second strategy to correct deviations in PLL-based commutation. A sensorless drive method based on the third harmonic of the back-EMF is introduced in [74] to drive BLDC motors and permanent magnet synchronous motors (PMSMs) in flux-weakening mode.

Fig. 12 depicts the block diagram of a BLDC motor drive system using a control method based on the virtual third harmonic of back-EMF signals. There are certain applications where accessing the neutral wire of the motor is not possible. However, the rotor position can be determined by comparing the voltage between the motor's neutral point 'N' and the virtual neutral point 'S'. According to [70], [75], [76], the virtual third harmonic back-EMF method is used to overcome this limitation. As part of this approach, the position signal is derived by measuring the voltage between the virtual neutral point 'S' and the midpoint of the DC-link 'P' (u_{SP}).

Voltage u_{SP} is calculated by

$$u_{SP} = \frac{u_{AG} + u_{BG} + u_{CG}}{3} - \frac{V_{dc}}{2} \quad (11)$$

where V_{dc} is the DC-bus voltage. As shown in [75], during the conduction periods of S_3 and S_4 , u_{SP} derives as follows:

$$u_{SP} = \frac{1}{3} \left(e_c - \frac{e_a + e_b}{2} \right) \quad (12)$$

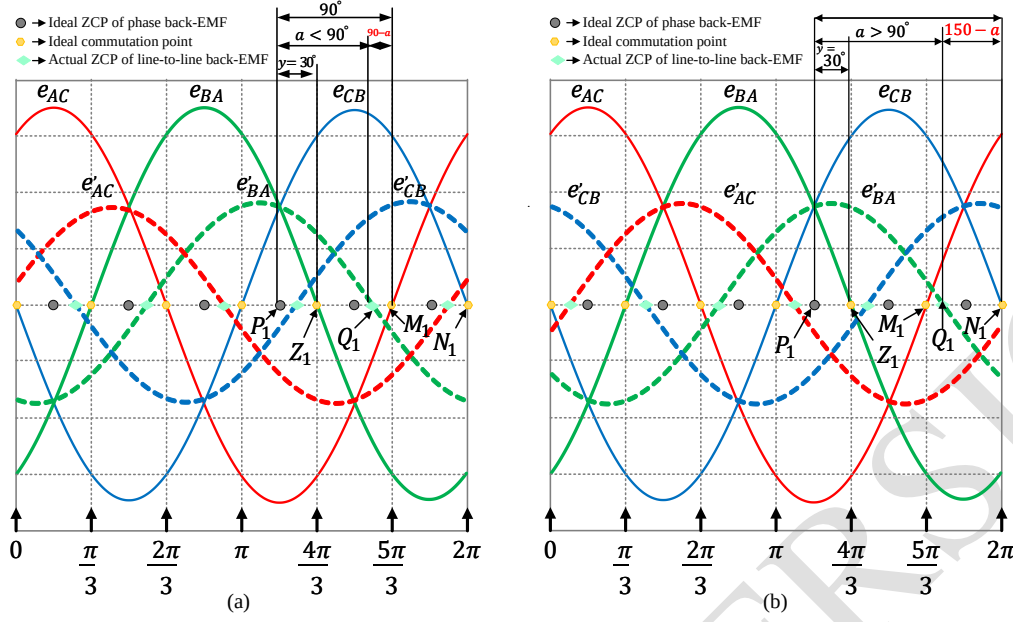


Fig. 8. Schematic diagram of the “90 – α ” commutation mode and the “150 – α ” commutation mode [57]. (a) “90 – α ” mode. (b) “150 – α ” mode.

The frequency of u_{SP} is three times the phase back-EMF frequency, which is equal to the back-EMF third harmonic frequency. Therefore, u_{SP} and back-EMF third harmonic are defined by the same ZCPs, as shown in Fig. 13. Based on the u_{SP} signal, a synchronic-frequency filter employing the second-order generalized integrator-based phase-locked loop (SFF-SOGI-PLL) is proposed in [75] to detect the rotor position of BLDC motors.

As discussed in [77], a sensorless drive technique can be implemented by using third harmonic flux linkage. As illustrated in Fig. 14, the flux linkages are generated by integrating the third harmonic back-EMFs. An alternative to using zero crossings of the third harmonic flux linkage is to use a continuous sinusoidal signal as a reference and accurate speed estimation errors are corrected at every sampling interval. It is common to use these methods for controlling motors that operate at low speeds. Nevertheless, their applications are

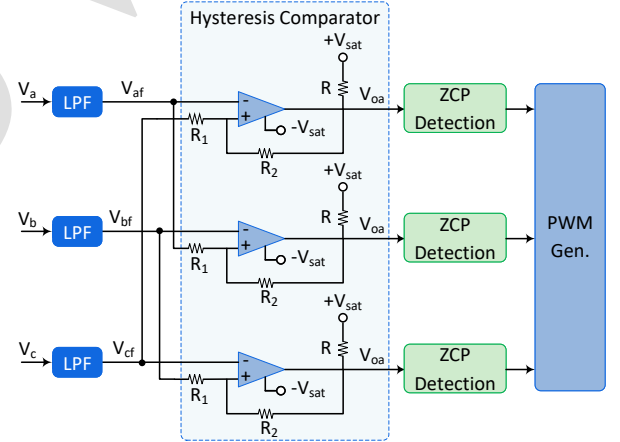


Fig. 10. Block diagram of sensorless control by using a hysteresis comparator [65].

limited due to the high level of interference [77].

C. Sensorless control method based on back-EMF integration (Back-EMF based)

Back-EMF integration has been proposed for detecting the position of BLDC motors in [78]–[82]. In this method, the back-EMF of the unexcited phase is integrated. The problem, however, is that errors can accumulate when applied to low-speed motors. In another sensorless method, the commutation signals can be generated using back-EMF integration and a PLL. A PLL is used in this method to maintain the integral result at zero. Its application is limited, however, by heavy

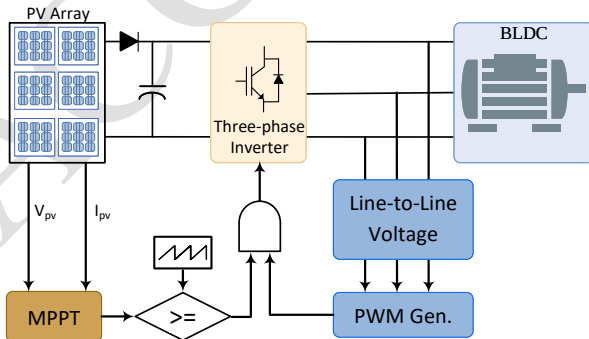


Fig. 9. Sensorless control method for a PV-based water pump [63].

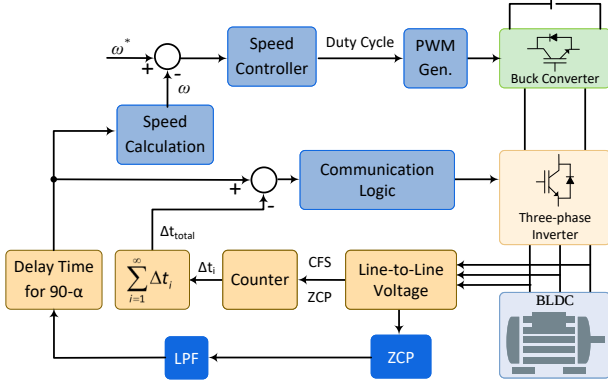


Fig. 11. Commutation error compensation control block diagram using $90-\alpha$ commutation mode [69].

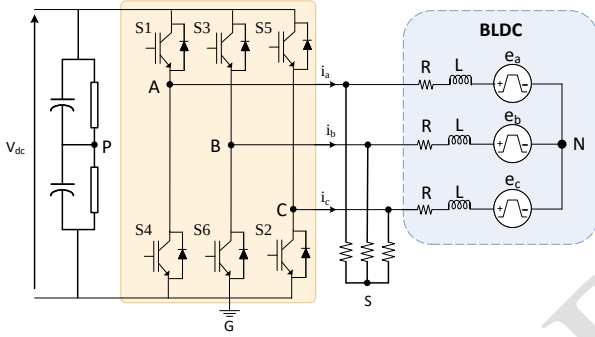


Fig. 12. Diagram of the drive system using a sensorless method based on virtual third harmonic voltage [75].

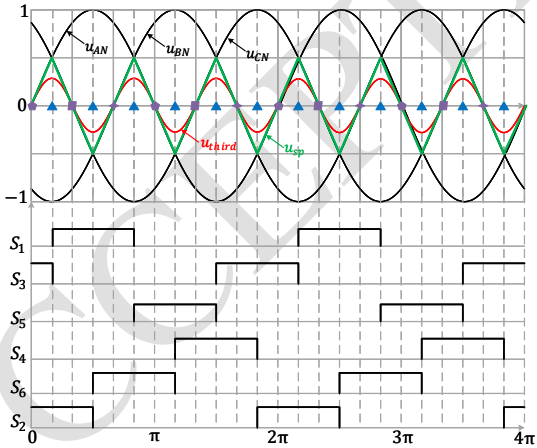


Fig. 13. Relationships between the phase back-EMFs, the third harmonic back-EMF, voltage u_{SP} , and the commutation points [75].

loads and high-speed commutation errors that can distort terminal voltages. Commutation errors are compensated by integrating adjacent 60-degree intervals before and after a commutation in order to address this issue [83].

Another integral prediction technique based on back-EMF

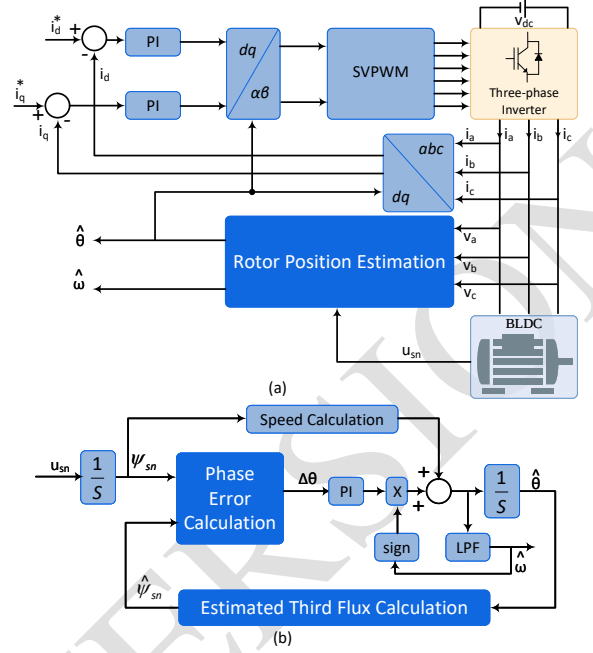


Fig. 14. Schematic of the control method using back-EMF third harmonic in [77] (a) Overall scheme (b) Rotor position estimator.

was introduced in [84]. This method has led to enhancements in commutation accuracy through the incorporation of two key improvements. Firstly, integral thresholds are dynamically adjusted at various speeds, considering the impact of LPFs and system delays. Secondly, an integral prediction method is employed to forecast the commutation instant, thereby mitigating errors arising from inadequate sampling during commutation [84].

D. Sensorless control method based on fuzzy control and neural network

Sensorless control of BLDC motors using a fuzzy control algorithm is proposed [85], [86]. A direct torque control (DTC) of BLDC motors based on fuzzy logic control is proposed in [87]. The fuzzification process involves mapping inputs of the control system (e_T , the error of estimated torque and $\hat{\theta}$, the estimated flux linkage angle) to their corresponding fuzzy sets within the input universe of discourse. In this case, the Output of the control system is space voltage vector (V_i , $i = 1, 2, \dots, 6$).

In this method, e_T is divided into four following fuzzy subsets:

$$\begin{cases} PB = \text{Positive Big} \\ PS = \text{Positive Small} \\ NB = \text{Negative Big} \\ NS = \text{Negative Small} \end{cases} \quad (13)$$

Also, flux linkage angle ($\hat{\theta}$) is divided into six fuzzy subsets $\{S_1, S_2, S_3, S_4, S_5, S_6\}$ and the rules of the Fuzzy logic controller has been provided in Table I [87]. The block

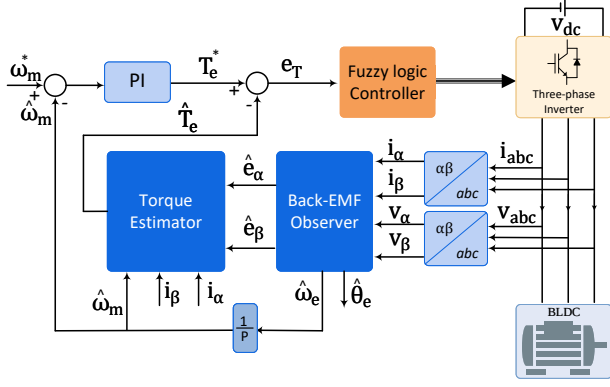


Fig. 15. Sensorless drive of a BLDC motor using fuzzy DTC method [87].

diagram of the sensorless drive of a BLDC motor using fuzzy DTC method is depicted in Fig. 15. The method uses a back-EMF observer that provides us with the estimation of back-EMF signals, estimated motor speed, and estimation of motor torque. Then, the fuzzy controller produces the switching signals based on the error of estimated torque. The proposed method does not perform well in low-speed range due to usage of the back-EMF observer.

TABLE I
FUZZY LOGIC RULES FOR DTC OF BLDC MOTOR [87].

$T_e \backslash \hat{\theta}$	S_1	S_2	S_3	S_4	S_5	S_6
PB	V_2	V_3	V_4	V_5	V_6	V_1
PS	V_2	V_3	V_4	V_5	V_6	V_1
NB	V_5	V_6	V_1	V_2	V_3	V_4
NS	V_5	V_6	V_1	V_2	V_3	V_4

Although it has a better dynamic response than linear proportional-integral (PI) controllers, it is more difficult to design and implement. The position of the rotor can be estimated using the freewheeling current in the unexcited phase [88]. Nevertheless, it requires complex hardware, which makes it challenging to use. The authors in [89] have proposed a fuzzy model to estimate the error of the estimated position of the motor, as well as to compensate for it. The estimation algorithm for a high-speed motor control scheme is, however, rather complex. A sensorless fuzzy logic commutation control scheme using equal area criteria for BLDC motor speed control is presented in [90].

In order to increase detection accuracy, sensorless detection methods based on neural networks [91], [92] have been developed. A neural network fuzzy controller, combined with the symmetry of phase voltage, is proposed in [93] for commutation deviation correction. Its performance is limited due to extensive computations, hardware/software requirements, and limited real-time capabilities. A sensorless neuro-fuzzy inference system (ANFIS) which is based on a multisector space vector PWM method (MS-SVPWM) is presented in [94]. As a result of this method, DC-bus voltage ripple, torque ripple, and total harmonic distortion are reduced. Furthermore, it reduces switching losses, which increases system efficiency.

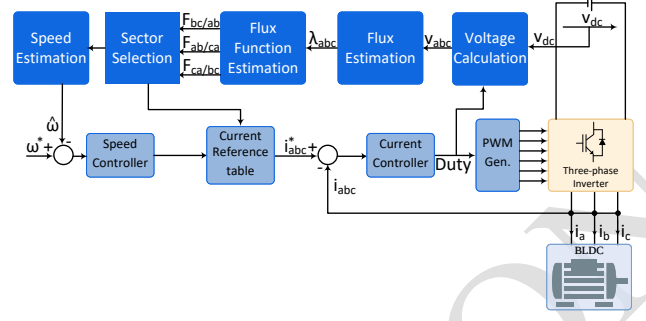


Fig. 16. Sensorless control method for the BLDC motor using flux-linkage function [100].

A sensorless method using phase-error correction algorithm based on a neuro-fuzzy controller for BLDC motor is presented in [95]. Authors in [96] have proposed a sensorless start-up current control method based on adaptive neural network inverse for high-speed BLDC motors. However, the method is complex to implement.

E. Sensorless control method based on PM flux linkage (Estimation and model-based method)

Sensorless control of PM motors based on flux-linkage estimation method is proposed in [97]–[99]. Due to its complexity, sensitivity to motor parameters, and error accumulation at low speed, this method is not suitable for all applications. A speed-independent flux linkage function has been proposed in [100] to estimate the commutation points of BLDC motors using the G-function method [101], as illustrated in Fig. 16.

Taking the BLDC motor equations into account, the line-to-line PM flux linkage of the motors can be derived as:

$$\begin{cases} \lambda_{ab} = \int_0^t [(V_a - V_b) - R(i_a - i_b)] dt - L(i_a - i_b) \\ \lambda_{bc} = \int_0^t [(V_b - V_c) - R(i_b - i_c)] dt - L(i_b - i_c) \\ \lambda_{ca} = \int_0^t [(V_c - V_a) - R(i_c - i_a)] dt - L(i_c - i_a) \end{cases} \quad (14)$$

To obtain the commutation point for phase a, the ZCP of line-to-line flux linkage λ_{bc} needs to be delayed by 30 electrical degrees. In this method, the ratio of line-to-line flux linkage is proposed as the flux-linkage function. For instance,

$$F_{ab/bc} = \frac{\lambda_{ab}}{\lambda_{bc}} \quad (15)$$

Commutation points can be detected by using flux-linkage functions, which are typically small, but increase significantly near extreme points. As shown in Fig. 17, the jumping edge of the flux-linkage function aligns with the line-to-line flux linkage ZCPs. Therefore, by delaying these jumping edges by 30 electrical degrees, it is possible to determine the commutation points.

For single-phase BLDC motors, a motion sensorless control technique is developed in [102]. It employs real-time estimation of PM flux and an initial current-frequency (I-F) sequence. A sensorless control method is introduced in [103], which

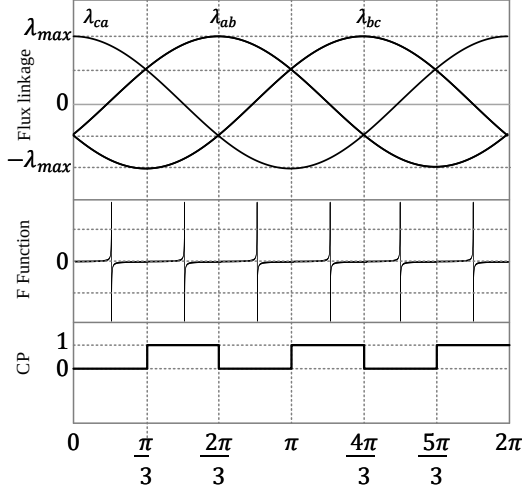


Fig. 17. Relation between line-to-line flux linkage signals, flux-linkage function, and commutation points [100].

relies on a speed-independent flux linkage function. In addition to enhancing the performance of BLDC motors during low-speed operation, this control approach also addresses phase discrepancies at high-speed operation.

In [104], a sensorless control method for BLDC motors is presented based on line-to-line flux linkage theory. A sensorless direct torque control (DTC) approach is described in [105]. In this strategy, the rotor position is estimated by using arctangent operations based on PM flux linkage of the motor in $\alpha\beta$ -frame. Based on line-to-line PM flux linkage, a finite element method (FEM)-based position observer for motion sensorless control of BLDC motors has been developed in [106].

F. Sensorless control methods using stochastic filters and estimation approaches (Estimation and model-based method)

Observer controllers and estimation approaches have also been proposed for sensorless control methods [107]–[116]. Integration of stochastic filters into control systems can be used to handle uncertainties within the system as well as external noise. In [109], a sensorless method is described utilizing back-EMF mapping, while in [110], torque constants are employed as the reference for a sensorless method. For sensorless control of brushless motors, a sliding mode observer (SMO) scheme can be utilized [111], [112]. It is proposed to replace the LPF with a more advanced Second Order Generalized Integrator (SOGI) in [117]. In order to obtain more accurate position estimation, [118] utilizes a SMO with Dual SOGIs and a Frequency Locked Loop (FLL).

A conventional SMO often suffers from chattering and distorted estimated back-EMF due to inadequate filtering [119], [120]. The use of high-order SMOs is preferred when it comes to reducing chattering and improving convergence time [121]. In [122], a second-order SMO with online parameter estimation is presented that considers temperature variations in stator resistance, which results in a higher level of stability.

A Dual PLL control system is also integrated with [123] to compensate for commutation errors during operation.

Observer-based sensorless control methods such as KF are also used for control of BLDC motors [113], [114]. Despite speed control being achieved, an everlasting error still exists when the rotor position is estimated. Reference [124] describes a sensorless direct torque control method for BLDC motors using KF. Based on [124], the Kalman filter estimates the state x_k :

$$\begin{cases} x_k = Ax_{k-1} + Bu_k + \omega_k \\ z_k = Hx_{k-1} + v_k \end{cases} \quad (16)$$

where A is the state matrix, B is the control matrix, H is the output matrix, x_k is the present state matrix, x_{k-1} is the previous state matrix, u_k is the input state matrix, z_k is the output state matrix and ω_k and v_k are process and measurement variables. Voltage vectors are selected based on torque and flux error to perform switching. The torque and flux errors are determined by comparing the actual and estimated values determined by the KF. The flux is derived from the motor's mathematical model. However, the estimated torque is highly inaccurate since the conventional KF model can only be applied to linear systems. Thus, to solve the mentioned problem, other types of KF control, such as Extended Kalman Filter (EKF) [115], [116], [125], [126], Unscented Kalman Filter (UKF) [127], and Cubature Kalman Filter (CKF) [128], have been developed.

The Particle Filter (PF) method is introduced in [129], [130] as an approach for highly nonlinear systems. In particular, this method is intended to estimate the position and speed of BLDC motors. An approach for nonlinear Model Predictive Control (MPC) for BLDC motors is presented in [131] which incorporates online state estimation techniques. In this study, three methods are evaluated: EKF, UKF, and the Nonlinear Moving Horizon Estimation (NMHE). According to [131], NMHE, when combined with nonlinear MPC, performs better than the other estimation methods. However, NMHE has a higher computational cost than EKF and UKF.

Additionally, [132] presents a sensorless method based on back-EMF observers. Trapezoidal back-EMFs are modeled as unknown inputs, and a proposed unknown input observer allows for real-time estimation of line-to-line back-EMFs, as depicted in Fig. 18. This method allows computation of rotor position without the need for additional circuits or complex operations. More back-EMF observers with phase-shift correction have been proposed in [133], [134].

G. Other novel sensorless control methods for BLDC motors

As presented in [135], the truth table can be used to obtain rotor position signals. Due to the elimination of switching harmonics, it has the advantage of removing the LPF, but it is not suitable for real-time applications. According to the study in [136], the phase self-inductances and resistances of the system exhibit minor fluctuations in response to rotor position. The B-H hysteresis characteristic of iron is responsible for these variations. This subtle change has been used to determine rotor position by introducing a measurement setup. The paper

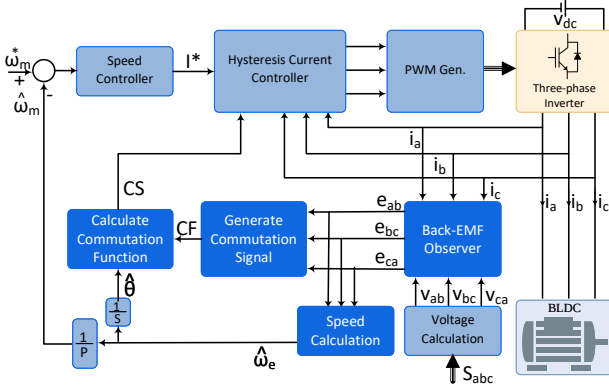


Fig. 18. Sensorless control based on back-EMF observer [132].

[137] describes a method to identify the position of a BLDC motor rotor and perform smooth motor startup without position sensors. It relies on detecting and utilizing the variation in inductance caused by rotor rotation.

A modified field-oriented control (FOC) for BLDCs is discussed in [138] as well as compared with conventional FOC. DTC method in contrast to FOC, responds faster to changes in torque or speed requirements, but it requires more computational burden [139], [140]. Hybrid controllers integrate a strategy that enables seamless switching between FOC and DTC [141]. FOC is efficient when operating in steady-state conditions, while DTC is efficient in situations involving transients [142].

It has been proposed in [143], [144] that a method based on DC-link current can be used to compensate for commutation errors. It allows a wide range of speeds to be achieved. The performance of the system in real-time is, however, limited. An adaptive control method for commutation error compensation was developed in [145] based on an analysis of the relationship between commutation error and dc-link current. The angle adjustment is performed by utilizing the symmetry of the dc-link current before and after commutation points. The system is also integrated with Model-Free Adaptive Control (MFAC) theory to deal with the nonlinearity of the phase back-EMF. The DC-link current integration method is proposed for low-inductance high-speed BLDC motors to compensate for commutation angle deviation [146]. In this method, commutation correction is performed by analyzing the integral difference between the DC-link current before and after commutation and deviation from the commutation angle, as shown in Fig. 19.

A method called the twelve-pulse production strategy is introduced in [147]. It creates two types of pulse with calculated time intervals between them. The Index Pulse (IP) corrects errors in detecting ZCPs and ensures an accurate 30-degree delay after ZCP detection to provide precise commutation timing. Another sensorless control method based on injecting higher time-harmonic components into the input voltages of the motor is proposed in [148]. However, this method reduces the system efficiency.

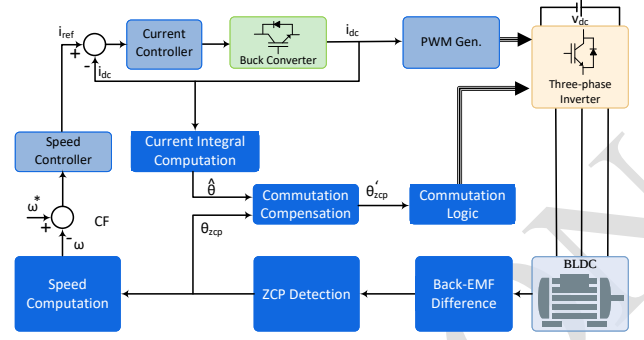


Fig. 19. Sensorless control block diagram of BLDC motor using DC-link integral method [146].

V. SENSORLESS CONTROL FOR BLDC MOTORS AT LOW SPEEDS AND START-UP

One of the current challenges of sensorless control of BLDC motors is the performance of the control system during start-up and the low-speed range [56]. Detecting ZCP accurately becomes more challenging at low speed due to small magnitude of back-EMF voltages. Consequently, its relative magnitude to measured noise is considerably lower compared to higher-speed scenarios. Also, the back-EMF integration method has the problem of accumulation of errors at low speeds. Furthermore, the virtual third harmonic back-EMF method is not suitable for very low speed range the substantial presence of noise and integral errors [56]. In addition, the freewheeling diode method detects the ZCPs by monitoring the on-off status of the freewheeling diode in the unexcited phase near the ZCPs. However, this approach necessitates additional circuits and involves a complex control mode [149].

Obtaining phase back-EMF based on a model-based method using mathematical model of the system introduces dynamic and inherent errors, leading to uncertain deviation of ZCPs. Furthermore, ZCP detection methods based on the F function and G function face challenges when it comes to ZCP detection, primarily due to the influence of the use of LPF and inherent limitations of mathematical models [56].

In [101], a sensorless approach utilizing a speed-independent function is introduced. This method can estimate commutation instants across a wide speed range, from near zero (2% of the rated speed) to high speeds. It's important to note that this technique is designed for surface-mounted permanent magnet BLDC motors [150]. The inductance method has proposed for low-speed control of BLDC motors by magnetizing or demagnetizing the winding using the saturation effect of iron core and the changing directions of the magnetic field from the permanent magnet [149]. The stator inductance changes with the rotor position because of saturation induced by the rotor magnet. Similarly, the rate of current change in the stator winding varies when a constant voltage is applied to the windings. This rate of current change can be used to estimate the relative position between a rotor magnet and a stator winding [151]. Another inductance-based method is proposed in [150]. This method involves injecting voltage pulses for

measuring inductance, and the accuracy of rotor detection is enhanced by measuring the peak inductance current. It has been shown that the method works from 0.5 % of rated speed (8 rpm). However, this technique is limited to salient pole BLDC motors, and its practical application is challenging [149].

Methods such as high-frequency injection is commonly applied to BLDC motors with a salient pole structure. In this method, high-frequency voltage (or current) signals are injected. This induces a response change through the spatial salient pole effect of the motor, allowing the calculation of rotor angle information by detecting this change [149].

The start-up methods in electric motors are commonly categorized into three types [152], [153]. The first approach is the open-loop start-up, also known as "align and go," where the rotor position is predetermined [154], [155]. This is accomplished by energizing two phases of the three-phase windings for a predetermined period. The permanent magnet rotor then adjusts its position to align with a particular angle. In order to initiate motor acceleration from a stationary state, an open-loop control scheme uses the known initial rotor position and a predetermined commutation logic. Even though this method is useful in specific automotive applications, it is associated with drawbacks, such as causing temporary vibrations and creating a substantial peak current [156]. Also, methods including V/f and I-f start-up techniques have been proposed for BLDC motor startup [157], [158]. An improved I-f start-up method with combining a closed-loop I-f control with an adaptive ramping current profile is proposed in [153] to deal with the issue of large start-up current caused by conventional open-loop startup method.

The second method involves utilizing inductance variation techniques during start-up [159], [160]. As discussed previously, in this method, when a stator winding aligns closely with the rotor's magnetic pole, the rate of current change within the stator winding, flowing in the direction of magnetization, is significantly higher than that in the opposite direction due to the saturation of the stator core by the magnetic field. Consequently, if a constant PWM voltage vector is applied to the motor's stator winding for a fixed duration of time, the current signal can be used to determine the rotor position [151].

Different voltage vectors for a three-phase voltage source inverter (VSI) is depicted in Fig. 20. Throughout the startup phase, the voltage vector in alignment with the identified q-axis is applied to optimize the q-axis current. This is achieved by continuously monitoring the rotor position using the mentioned method and subsequently applying an appropriate voltage vector. To monitor the rotor position two short detection voltage vectors are applied after each acceleration cycle before injecting the drive voltage vector. These two detection voltage vectors determine whether the new rotor position aligns with the expected sector [151], [160].

The third strategy employs a KF to estimate rotor position based on measured voltage and current [153]. However, it is possible to produce substantial instantaneous currents with the second and third types when large voltage pulses are injected. Not only does this impose higher specifications on power

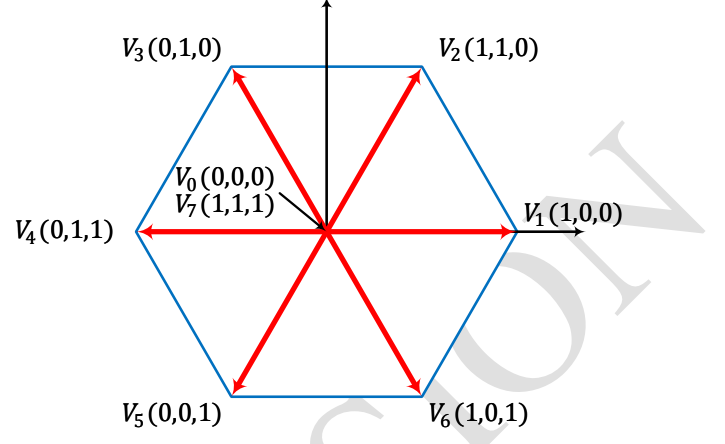


Fig. 20. Possible voltage vectors for a VSI.

switches and increase hardware circuit costs, it also introduces the risk of excessive acceleration, leading to potentially failed startup [153], [156].

VI. SENSORLESS CONTROL FOR BLDC MOTORS WITH NONIDEAL BACK-EMF

In many applications especially the high-speed BLDC motors with low-inductance, the back-EMF signals are not ideal trapezoidal signals as depicted in Fig. 21 [161], [162].

The commutation affects the shape of the back-EMF signals [163]. For instance, when the current transfers from phase A to phase B, the current in phase A doesn't reach to zero instantly. It decreases gradually and during the commutation period, it flows through the freewheeling diode. During the

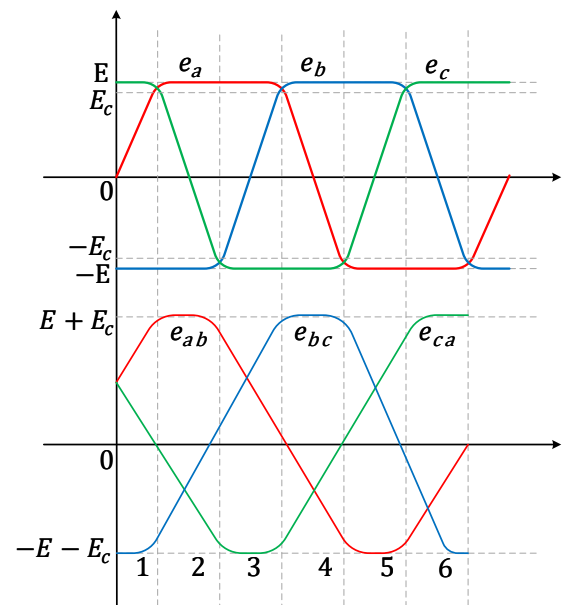


Fig. 21. Nonideal back-EMF voltages of a high-speed BLDC motor.

commutation period, $V_A = 0$, $V_B = E$, and $V_C = 0$. There for the differential equations for the stator currents are as follows:

$$\begin{cases} Ri_A + L \frac{di_A}{dt} = \frac{e_B + e_C - 2e_A - E}{3} = v_{A,com} \\ Ri_B + L \frac{di_B}{dt} = \frac{2E + e_A + e_C - 2e_B}{3} = v_{B,com} \\ Ri_C + L \frac{di_C}{dt} = \frac{e_B + e_A - 2e_C - E}{3} = v_{C,com} \end{cases} \quad (17)$$

Considering the initial conditions for the phase currents i.e. $i_{A,0} = I_{dc}$, $i_{B,0} = 0$, $i_{C,0} = -I_{dc}$, the phase currents during the commutation period will be

$$\begin{cases} i_A = \frac{v_{A,com}}{R} + (I_{dc} - \frac{v_{A,com}}{R})e^{-\frac{R}{L}t} \\ i_B = \frac{v_{B,com}}{R} - \frac{v_{B,com}}{R}e^{-\frac{R}{L}t} \\ i_C = \frac{v_{C,com}}{R} - (I_{dc} + \frac{v_{C,com}}{R})e^{-\frac{R}{L}t} \end{cases} \quad (18)$$

Equation (18) results in not accurate commutation points especially in medium and high-speed applications that should be compensated [163]. An extended estate observer (ESO) method is proposed in [164] to estimate the back-EMF signals of the BLDC motor. The commutation error is subsequently derived from this estimated back-EMF. Unlike the traditional closed-loop correction method based on back-EMF symmetry, the ESO-based method suggested for compensating commutation error achieves error determination within a single commutation cycle.

VII. BLDC MOTOR DRIVE IN PRESENCE OF ASYMMETRIC BACK-EMF

The asymmetry of motor parameters such as stator resistance and inductance affects sensorless commutation signals. In addition to motors with asymmetric parameters, unbalanced ZCP detection can also be caused by DC-link voltage drops or DC-link offset voltages. As shown in Fig. 22, ZCPs of sensorless methods are deviated from their ideal positions when asymmetric parameters are present. Therefore, signals generated by the commutation system are inaccurate, resulting in problems with the control system. It may also result in increased ripples in torque and current. Consequently, conventional sensorless control of BLDC motors with unbalanced ZCP detection may cause instability [161], [165], [166].

A sensorless drive for BLDC motors based on the third harmonic back-EMF signals is described in [165]. In contrast to conventional methods, analog-to-digital converters do not use LPFs and, as a result, there is no phase lag. A dynamic oversampling technique is employed in high-speed regions to reduce ZCP detection sampling delay. Furthermore, a method for compensating for commutation errors caused by asymmetric back-EMF is proposed in [165], as shown in Fig. 23.

A linearization method based on input-output feedback is described in [167]. The linearized model of the system is then used to describe a sensorless optimal commutation control method for steady-state operation. The method is

proposed for correcting commutation deviations using one current sensor while considering asymmetric back-EMF [168]. A dual improved second-order generalized integrator with a positive sequence component extractor (DISOGI-PSCE) is described in [169]. This deviation correction method takes into account back-EMF asymmetry of BLDC motors. Additionally, [170] proposes a current integral method that addresses the challenges associated with asymmetric back-EMFs and system robustness.

VIII. DISCUSSION

The terminal voltage sensing (ZCP detection) method is a straightforward approach widely used in various industrial applications. However, its reliance on a LPF introduces a significant phase lag at high-speed applications. To address this issue, recent advancements, such as employing the '150 - α ' commutation mode, have been proposed. Therefore, the speed of 20000 rpm has been investigated in [57] based on the aforementioned method. On the other hand, the application of this method is limited at low speeds due to the small magnitude of back-EMF voltage in this range. Typically, this method have been used in the speed range of 1000 to 6000 rpm [39].

Methods based on third harmonic back-EMF signals have been used in application with wide speed range. Their resilience against noise and ease of signal detection make them particularly effective in the lower speed range compared to terminal voltage sensing methods. Since these signals have three times the frequency of fundamental back-EMF voltages, they can be used for lower speed ranges, such as 100 to 6000 rpm, as discussed in [39]. It has been discussed that methods based on third harmonic back-EMF signals cannot be applied to very low-speed ranges, and several recent studies have attempted to improve this method. As a recent advancement, a virtual third harmonic method is used for BLDC motors with asymmetric back-EMF signals for high-speed applications (around 60000 rpm) [165].

Back-EMF integration method has the advantage of attenuating high-frequency noise. However, a much higher sampling rate than the commutation frequency is required, which limits its application at very high speed. Thus, the insufficient sampling signal points can lead to an inaccurate commutation. As it is discussed earlier this method have limitations in low-speed range. Recent research has been done to mitigate the issues of this method for high-speed applications as 15000 rpm has been investigated in [81]. Flux linkage method is more robust against noise and disturbances which is the reason that this method has been used for low-speed applications [101]. Additionally, a method based on speed-independent flux linkage is introduced in [103] for a speed range spanning from 1% of full speed up to 20000 rpm.

Fuzzy logic and neural network control methods have the potential to enhance the accuracy and performance of control systems. Nevertheless, their application faces challenges due to computational burden, demand for significant calculation resources, and potential software delays, posing practical challenges. Furthermore, in specific applications, these methods may depend on back-EMF information, which makes them

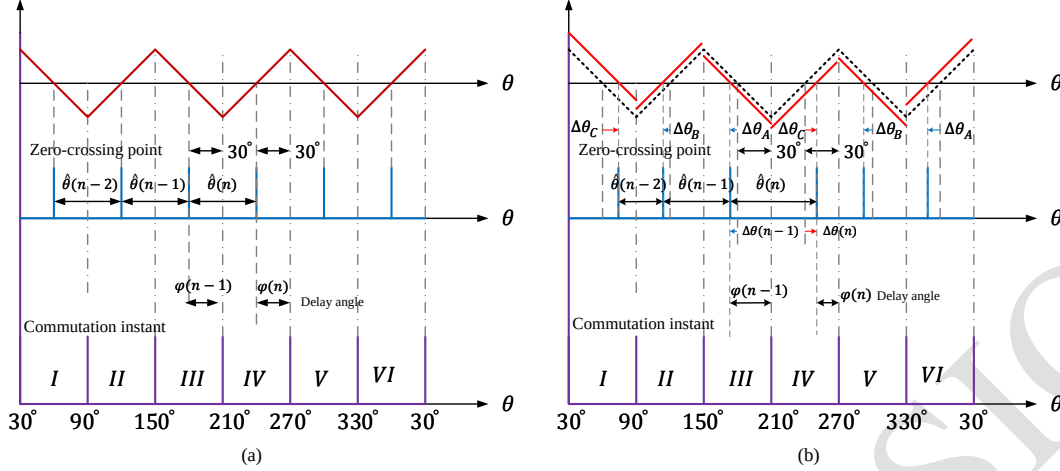


Fig. 22. Comparison of ZCP interval $\hat{\theta}(n)$, ZCP error $\Delta\theta(n)$, and commutation delay angle $\varphi(n)$ for symmetrical and asymmetric back-EMF signals [165]. (a) Symmetrical impedances. (b) Asymmetric impedances.

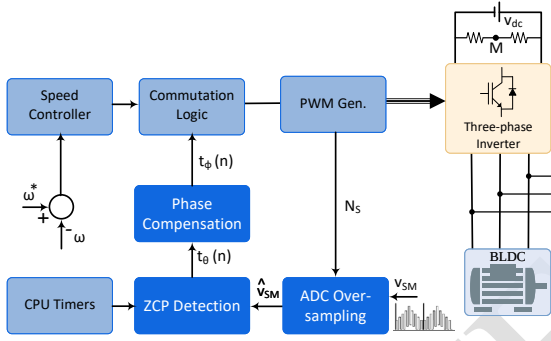


Fig. 23. Sensorless control of BLDC motor with commutation phase compensation using analog to digital converter [165].

sensitive to the accuracy of the detection or estimation method. An example of a high-speed BLDC motor, controlled by the fuzzy logic method, reaching up to 15000 rpm, is explored in [171].

Different types of stochastic filters have gained attention in practical and industrial applications due to their ability to reject disturbances, noise, and uncertainty in model parameters [39]. As a result of their ability to reject noise and disturbance, these methods are also used in low-speed ranges [124]. The low-speed range, approximately 10-20 rpm up to higher speeds (around 2000 rpm), has been explored for EKF and CKF in [128]. It should be noted, however, that these methods have some limitations when it comes to practical applications. It is necessary to consider the computational burden, the memory storage requirements, and the computing constraints. It may be possible to mitigate the practical concerns with this kind of method in the future with advancements in processor technology.

To summarize, different sensorless control methods for BLDC motor drive application are summarized in Table II. There are various research directions to tackle challenges in

sensorless control of BLDC motors. A lot of effort has been done on improving the detection or estimation of trapezoidal back-EMF voltages and in particular, their ZCPs. However, achieving high-performance sensorless control in high-speed applications is difficult due to potential phase lag of LPF, software delays, and system delays in many control methods. On the other hand, sensorless methods relying on back-EMF voltage detection or estimation often struggle in the low-speed range and startup processes, prompting research in these areas. It is crucial when it comes to the development of self-starting methods for BLDC motors under standstill conditions. Challenges in this domain include limiting startup current, addressing issues like reverse torque and startup delays, and ensuring smooth operation from zero to high speeds. Another potential research area is minimizing torque ripple, especially in high-speed applications. Furthermore, considering the impact of nonideal or asymmetric back-EMF voltages in the design of control systems is a recent and significant topic in sensorless control of BLDC motors, which is essential for real-world industrial applications.

IX. CONCLUSION

The purpose of this paper is to review the sensorless control methods for BLDC motor drives. The benefits, limitations, challenges, and applications of a variety of sensorless methods are discussed and reviewed, including detecting ZCPs with terminal voltage measurement, third-harmonic back-EMF voltage, back-EMF integration, PM flux linkage, fuzzy control, observers, and stochastic filters. Sensorless techniques relying on terminal voltage sensing are relatively straightforward to implement. It is necessary, however, to use a LPF to detect ZCPs which presents challenges, particularly in high-speed applications. The LPF can introduce significant phase delays in such scenarios, which must be compensated. Several research efforts have been conducted to enhance ZCP detection for improving sensorless control for high-speed BLDC motors. These efforts include eliminating the need for an LPF or

TABLE II
COMPARISON OF DIFFERENT SENSORLESS CONTROL METHODS FOR BLDC MOTOR DRIVE.

Method	Technique	Advantages	Disadvantages	References
Terminal voltage sensing	ZCP detection based, phase shift compensation	Easy to implement	Difficult ZCP detection at low speed, Phase lag error at high speed	[41]-[52] [56]-[61], [66]-[69]
Third harmonic approach	Virtual third harmonic, Third harmonic integration	Not affected by PWM noise, Higher cutoff frequency for LPF	Integral errors at low speed	[70]-[77]
Back-EMF integration	Integrating the back-EMF of the unexcited phase	Can be used for wider range of speed	Accumulation of errors at low speed, Terminal voltage distortions at high speed	[78]-[84]
Fuzzy control and neural network	Fuzzy logic controllers, Artificial neural network	Higher accuracy, Superior dynamic performance	Complex algorithms	[85]-[90] [91]-[96]
PM flux-linkage function	Estimation of PM flux linkage, G-function method	Not affected by noise, Speed-independent flux linkage function	Sensitivity to motor parameters, Error accumulation at low speeds	[97]-[104]
Stochastic filters	Different type of Kalman Filters, Model Predictive Control, Nonlinear Moving Horizon Estimation	Dealing with system uncertainties and external noise, Dealing with nonlinear systems	Computational complexity, Tuning challenge	[115]-[116] [125]-[132]
Observers	Back-EMF observers, Sliding mode observers	Can be used for wider range of speed	Sensitivity to motor parameters	[111]-[112] [117]-[123]

enhancing the approach for phase shift compensation. The relatively low amplitude of back-EMF signals at low speeds can also pose an obstacle to sensorless control of BLDC motors. A number of sensorless drive technologies have been introduced to address this issue, including those making use of third harmonic back-EMF signals or flux-linkage functions. To improve the accuracy of the control system, fuzzy control and neural networks have been proposed in some research papers. Nevertheless, these approaches can be complex to implement. Additionally, recent research that has addressed the issue of sensorless control considering the challenges of nonideal back-EMF signals which affect ZCP detection accuracy, or asymmetric back-EMF signals, is discussed in this paper.

ACKNOWLEDGEMENT

This work was carried out within the frame of a collaborative industrial/academic research project funded by SOGEFI Air & Cooling, France, SATT Conectus/University of Strasbourg, France, and University of Lorraine, France.

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